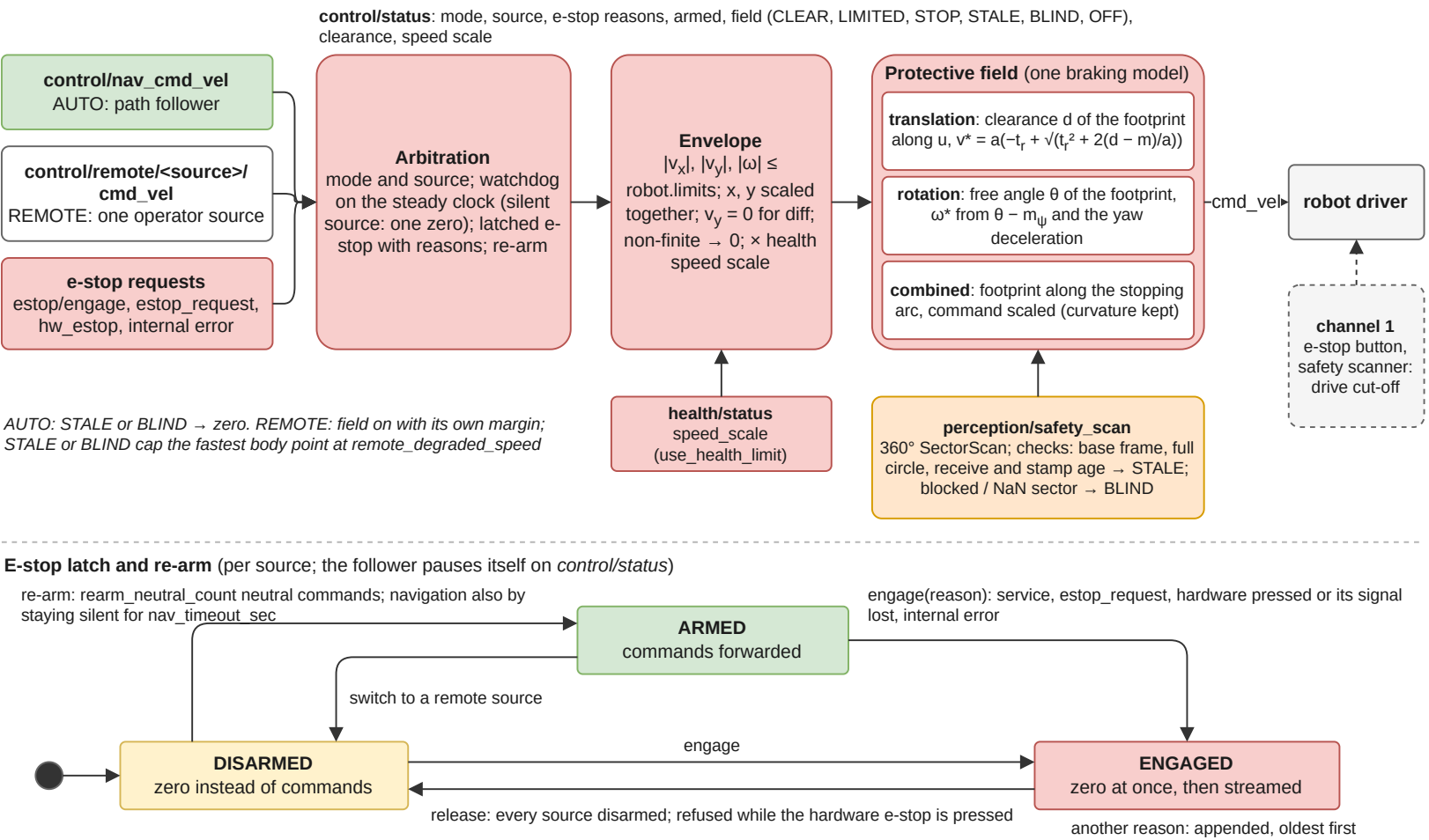
control/status: mode, source, e-stop reasons, armed, field (CLEAR, LIMITED, STOP, STALE, BLIND, OFF), clearance, speed scale
control/nav_cmd_vel
AUTO: path follower
control/remote/<source>/ cmd_vel
REMOTE: one operator source
e-stop requests
estop/engage, estop_request, hw_estop, internal error
Arbitration
mode and source; watchdog on the steady clock (silent source: one zero); latched e-stop with reasons; re-arm
Envelope
$|v_x|, |v_y|, |\omega| \leq$ robot.limits; x, y scaled together; $v_y = 0$ for diff; non-finite → 0; × health speed scale
Protective field (one braking model)
translation: clearance d of the footprint along u, $v^* = a(-t_r + \sqrt{(t_r^2 + 2(d - m)/a)})$
rotation: free angle θ of the footprint, ω^* from $\theta - m_\psi$ and the yaw deceleration
combined: footprint along the stopping arc, command scaled (curvature kept)
cmd_vel
robot driver
channel 1
e-stop button, safety scanner: drive cut-off
AUTO: STALE or BLIND → zero. REMOTE: field on with its own margin; STALE or BLIND cap the fastest body point at remote_degraded_speed
health/status
speed_scale
(use_health_limit)
perception/safety_scan
360° SectorScan; checks: base frame, full circle, receive and stamp age → STALE; blocked / NaN sector → BLIND
E-stop latch and re-arm (per source; the follower pauses itself on *control/status*)
re-arm: rearm_neutral_count neutral commands; navigation also by staying silent for nav_timeout_sec
engage(reason): service, estop_request, hardware pressed or its signal lost, internal error
ARMED
commands forwarded
switch to a remote source
DISARMED
zero instead of commands
engage
ENGAGED
zero at once, then streamed
release: every source disarmed; refused while the hardware e-stop is pressed
another reason: appended, oldest first